COEN 317 Assignment 2 Simon Gunidan 40058301

Question 1 (assuming 32 bits registers)

1.a) R1: 243 R2: -7777 b) N 0 Z 0 C 0 V 0 Q 0
2.a) R1: 243 R2: -7777 b) N 0 Z 0 C 0 V 0 Q 0
3.a) R1: 7777 R2: 134 217 484 b) N 0 Z 0 C 0 V 0 Q 0
4.a) R1: 7777 R2 134 217 484 b) N 0 Z 0 C 1 V 0 Q 0
5.a) R1: 243 R2: -7777 b) N 0 Z 0 C 0 V 0 Q 0
6.a) R1: 243 R2: -7777 b) N 0 Z 0 C 1 V 0 Q 0
7.a) R1: 7777 R2: -4084 b) N 0 Z 0 C 0 V 0 Q 0
E61 8.a) R1: 7777 R2: -4084 b) N 1 Z 0 C 1 V 0 Q 0

→ For 1. $(7777)_{10}$ → 0001 1110 0110 0001, shift right by 5 → 0000 0000 1111 0011 = $(243)_{10}$
For 2. same as 1. but we set flags, since no flags were set in 1., same answer
19F For 3. shift hexadecimal by 5 to the left: 0x07FFFF0C so $(134,217,484)_{10}$
Here, bit at position 4 was 1 so carry takes value 1 but instruction has no S
For 4. same as 3 but since our instruction has S, carry takes value 1
For 5, since bit 31 is 0 and instruction doesn't have S, same as 1.
For 6, since bit 31 is 0 and bit 4 is 1, same as 5. except carry is set to 1
For 7, Since bit 31 is 1 and bit 4 is 1, 1111 1111 1111 1111 1110 0001 1001 1111
1111 1111 1111 1111 1111 1110 0000 1100
so N=1 and C set to 1 BUT nos F F F F F F 0 C → -4084
For 8, Same as 7. but we set the flags this time

Question 2

a) $y = z + x/64$ ⇝ right shift x by 6 to divide by 64
ASR R1, R1, #7 (divide x by 64)
ADD R2, R3, R1 y = z + (new x)

b) $z = 8*z + y*x$ ⇝ shift z left by 3 (multiply by 8); shift y by x
LSL R3, R3, #3 } z*8
LSL R2, R2, R1 } y*x
ADD R3, R3, R2 } z = z+y

c) LSL R1, R1, 8 } 128x z = z - y → SUB R3, R3, R2
ASR R3, R3, 1 } z/2
ADD R3, R3, R1 z = z+x

Question 3

R1: A000 D4BA → -1 610 558 278
R2: B001 B1C2 → -1 342 066 238
R3: 7FFF FFFE → 2 147 483 646
R4: 8000 0002 → -2 147 483 646

	R1	R2	R3	R4	N	Z	C	V	Q
a)	5002 867C	B001 B1C2	7FFF FFFE	8000 0002	0	0	1	0	0
b)	A000 D4BA	B001 B1C2	5002 867C	8000 0002	0	0	1	0	0
c)	5002 867C	B001 B1C2	7FFF FFFE	8000 0002	0	0	1	1	0
d)	A000 D4BA	B001 B1C2	0000 0000	8000 0002	0	1	0	0	0
e)	A000 D4BA	B001 B1C2	5002 867D	8000 0002	0	0	1	1	0
f)	A000 D4BA	B001 B1C2	7FFF FFFE	8000 0002	0	0	0	0	0
g)	A000 D4BA	0000 0004	7FFF FFFE	8000 0002	0	0	1	1	0
h)	8000 0000	B001 B1C2	7FFF FFFE	8000 0002	0	0	0	0	0
i)	A000 D4BA	A000 D4BB	7FFF FFFE	8000 0002	0	0	0	0	0

Question 4

1) LDR R0, [R1, R2, LSL#0x6] shift R2 by 6 bits ⇒ 0x0012 → 0x0480

0000 0001 0010
0100 1000 0000

Then R1 + R2: 0x1F34 8796
+ 0x 0000 0480
0x 1F34 8C 16 ← memory being addressed

so R0 = 9898 9898 (hex) R1 = 1F34 8796 R2 → 0x 480

2) LDR R0, [R1, #0x7]! → Pre index addressing

Memory being addressed: R1 + 0x7 → 0x1F34 8796
7
0x 1F34 879D → memory being addressed

R0: 0x98989898
R1: 0x1F34 879D
R2 0x0000 0012

3) LDR R0, [R1], #0x8 → post addressing

memory addressed is value in R1 → 0x1F34 8796
R0: 0x98989898
R1: 0x1F34 879E (incremented after load)
R2: 0x 0000 0012

Question 5

	mem addressed	R0	R1	R2
① LDR R0, [R1, R2, LSL #0x6]	1F34 8C16	98989898	1F34 8796	0000
② LDR R0, [R1, #0x7]!	1F34 879D	98989898	1F34 879D	0000
③ LDR R0, [R1], #0x8	1F34 879D	9898 9898	1F34 87A5	0000

① ⇝ 1st step was same as problem 4, now we increment R1 by 0x7

② R1 = 1F34 8796
+ 0000 0007

0x1F34 879D → used in mem add.

③ First we address memory in R1, then add 0x8:

1F34 879D
0000 0008
1F34 87A5

Q6 a) First we convert 0x1100 to dec ⇝

0001 0001 0000 0000 = 4352

If we decrement R2 4352 time, we set the Z flag to 1 to leave the loop so we go through the BNE DLY instruction 4352 times

b) to only go through BNE DLY 99 time, we set R2 to 0x03 (0110 0011)

c) 4352 ÷ 99 = 43.95

if we decrement by 43 99 times we obtain 4257

4352 - 4257 = 95 so we need to replace NOP to deal with the leftover